

Ayanda Thabo Phaketsi

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Education

University of Cape Town

BSc Computer Science and Business Computing

Cape Town, South Africa

Dec 2025

Relevant Coursework: Software Engineering, Algorithms and Data Structures, Databases, Operating Systems, Computer Networks, Machine Learning, Systems Analysis and Design

Projects

clia – Autonomous Local Software Engineer AI Assistant

2025

<https://pypi.org/project/clia-swe-ai/>

Project Description: Developed and published an advanced conversational AI assistant designed to automate complex software engineering tasks. Operating as both a CLI and Web UI, it enables developers to refactor, debug, and implement features through natural language by directly interacting with the local developer environment.

Key Contributions:

- **Core AI Logic:** Implemented an asynchronous loop managing multi-turn exchanges with Google Gemini, handling streaming thoughts, and orchestrating complex tool usage.
- **Model Context Protocol (MCP):** Leveraged MCP to build a dedicated tool server that enables safe, deterministic filesystem and command-line interactions.
- **Safety & Transparency:** Engineered a "Human-in-the-loop" permission model, requiring explicit user approval for destructive operations while streaming "AI thoughts" for real-time plan visibility.
- **Advanced Tooling:** Built a suite of specialized tools for line-level code editing, codebase snapshots, and asynchronous process management.

Technologies Used:

- **Languages/Frameworks:** Python, Google Gemini API, FastAPI, Model Context Protocol (MCP)
- **UI Tools:** Rich (CLI aesthetics), prompt-toolkit (interactive terminal), WebSockets (persistent web chat)

MemGallery – AI-Powered Memory Assistant (Android)

2025

<https://github.com/Past-da-king/MemGallery>

Project Description: Designed a universal "external brain" application that democratizes brand-exclusive AI features (like Pixel Screenshot and Nothing Essential Space) for all Android 7.0+ devices. It automatically captures and processes various media types to extract actionable insights and enable semantic search.

Key Contributions:

- **Screenshot Detection Architecture:** Engineered a background system using ContentObservers and WorkManager to detect, filter, and process screenshots with zero manual input.
- **Multimodal Processing:** Integrated Gemini 2.5 Flash to analyze images, audio transcriptions, and text notes, extracting events, todos, and reminders into a relational Room database.
- **Semantic Search:** Implemented a searchable memory bank where users can find items using natural language descriptions (e.g., "that meeting about the coffee shop").
- **Premium UX/UI:** Built a modern Material Design 3 interface with dynamic color theming, shared element transitions, and high-performance staggered grid layouts.

Technologies Used:

- **Mobile Stack:** Kotlin, Jetpack Compose, Hilt DI, Room Database, Coroutines/Flow
- **System APIs:** WorkManager, ContentObservers, Google Generative AI SDK, MediaStore

Nexa – Decentralized Offline-First Communication Platform

2025

<https://github.com/Past-da-king/rednexa>

Project Description: Led a 3-person CS3 Capstone team to bridge the digital divide in rural communities by building a decentralized communication ecosystem. The app allows secure messaging and social networking in areas with zero internet access through a innovative peer-to-peer network.

Key Contributions:

- **Delay-Tolerant Networking (DTN):** Implemented a custom DTN protocol using "message hopping," allowing data to traverse the network via Bluetooth and Wi-Fi Direct relays.
- **Protocol Optimization:** Developed deduplication, Time-to-Live (TTL), and hop-count logic to prevent network flooding and ensure delivery resilience.
- **Layered Encryption:** Integrated Google Tink for end-to-end encryption of content and hop-by-hop encryption for DTN metadata, ensuring privacy even through relay nodes.
- **Offline-First Identity:** Created a system for local identity management (no phone number required), leveraging QR codes and proximity scanning for peer discovery.

Technologies Used:

- **Core Stack:** Kotlin, Google Nearby Connections API, Google Tink, Jetpack Compose
- **Data Management:** Room, Kotlinx Serialization, Hilt

StudyInc – Collaborative Productivity Suite

2025

<https://phaketsi.vercel.app/#projects>

Project Description: Architected a full-stack collaborative platform (Electron Desktop App) designed for intensive focus sessions. It integrates real-time peer presence, pomodoro-style sprints, and data-driven learning analytics to improve student productivity.

Key Contributions:

- **Real-time Syncing:** Leveraged Socket.io for instant chat, live timer synchronization, and WebRTC signaling for peer-to-peer video calls within study groups.
- **Desktop Experience:** Utilized Electron to implement native features like global "always-on-top" mini-overlays for timers and tasks, achieved through precise IPC management.
- **Analytics Dashboard:** Developed a dynamic dashboard using React and Tailwind CSS to visualize study habits, daily averages, and course-specific progress stats.
- **Scalable Backend:** Built a Node.js/Express.js server with Mongoose ODM for MongoDB, implementing secure JWT-based authentication with cross-origin socket support.

Technologies Used:

- **Tech Stack:** Node.js, Next.js, Electron, MongoDB, Socket.io, WebRTC
- **Styling:** Tailwind CSS, shadcn/ui, Radix UI

Verilookie – Unified Deepfake & AI Content Detection

2025

<https://github.com/Maphuti-Shilabje/verilookie>

Project Description: Developed a full-stack web application designed to combat misinformation by providing a multi-layered pipeline for detecting deepfakes and AI-generated media, combined with a gamified learning platform.

Key Contributions:

- **Unified Detection Pipeline:** Integrated Hugging Face Vision Transformers (ViT) and NVIDIA AI APIs into a sequential detection logic to achieve high-accuracy verdicts.
- **Gamified Education:** Designed a quiz system powered by Gemini API that dynamically generates adaptive-difficulty challenges based on the user's weighted accuracy history.
- **Performance Optimization:** Implemented backend frame sampling using OpenCV for efficient video analysis and containerized the entire suite using Docker.

Technologies Used:

- **Backend:** Python (FastAPI), PyTorch, OpenCV, Docker
- **Frontend/AI:** React, Google Gemini API, Hugging Face Transformers

Awards & Achievements

- **Winner – Entelect University Challenge 2025:** Recognized for technical excellence and competitive problem-solving (Featured in UCT Official News).
- **2nd Place – Predictive Insights 2025 Hackathon:** Developed a data-driven solution for predictive modeling.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, Kotlin, C#, SQL, $\text{\LaTeX}$

Frameworks/Tools: React, Next.js, FastAPI, Node.js, Jetpack Compose, Electron, MongoDB, Docker, Git

Specializations: AI/LLM Integration, Distributed Systems (DTN), Full-Stack Mobile & Web